



Long-term game species dynamic as indicator for changing landscape quality

Erich Tasser^a, Birgith Unterthurner^{a,b}, Andreas Agreiter^c, Heinrich Aukenthaler^b, Lothar Gerstgrasser^b, Marco Giardino^a, Ulrike Tappeiner^{a,d}, Johannes Rüdiger^{d,*}

^a Eurac Research, Institute of Alpine Environment, Bozen/Bolzano, Italy

^b South Tyrolian Hunting Association, Bozen/Bolzano, Italy

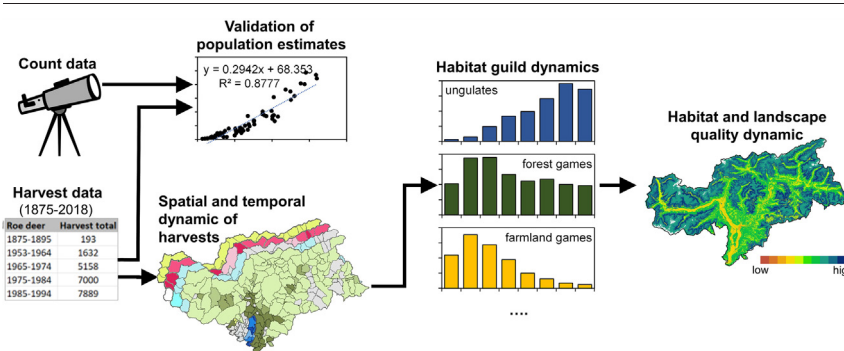
^c Office for Hunting and Fisheries, Autonomous Province of Bozen/Bolzano, Italy

^d Universität Innsbruck, Department of Ecology, Innsbruck, Austria

HIGHLIGHTS

- Hunting data were used to assess dynamics of species and habitat guilds since 1870.
- Spatio-temporal dynamics of 28 game species were analysed in South Tyrol, Italy.
- Ungulates profited from the hunting management and land-use changes.
- Habitat quality for farmland and synanthropic species strongly decreased.
- After initial increases in quality, forests and alpine regions lost quality.

GRAPHICAL ABSTRACT



ARTICLE INFO

Editor: Paulo Pereira

Keywords:

Agricultural intensification
Forestry
Habitat guild
Hunting
Population ecology
Specialists

ABSTRACT

Urban sprawl, increased traffic and modern forestry, as well as the globalisation of agriculture, have increasingly been affecting the landscape and its quality as habitat for species especially since the middle of the last century. Still, there are hardly any methods nor indicators which can measure the quality of the landscape for species over a long period. In this study, we investigated the influence of landscape structure and landscape quality on harvest data of 28 game species in South Tyrol, Italy, over the last 150 years. The harvest data were used to assess the population dynamics of individual species and habitat guilds since 1870.

As a first result, we could show, on the examples of six species, that count population data were highly correlated with harvest data and are therefore well suited to estimate their population size. Second, the populations of ungulates consistently increased during the study period. The numbers of mesocarnivores as well as smaller forest and alpine game species increased strongly until the 1970s/80s of the last century, followed by a decline. The populations of farmland species and some synanthropic species have decreased substantially, and some species have even disappeared completely. Based on these results, we were able to show, in a third step, that the landscape quality for game species in South Tyrol has developed differently: In particular, the agriculturally used habitats have lost quality, whereas forests and alpine regions have initially gained quality due to the extensification of use; during the last five decades, the quality decreases again, at least for small game species. Our results thus provide concrete implications for the active improvement of the landscape quality for farmland and forest species, as well as indications for future priorities in funding support of alpine pasture management.

* Corresponding author at: Sternwartestraße 15, 6020 Innsbruck, Austria
E-mail address: Johannes.Ruediger@uibk.ac.at (J. Rüdiger).

1. Introduction

Game species are excellent indicator species for environmental changes (Hanley, 1996). This is because (1) their biology and ecology are generally well known; (2) they have relatively large rather than isolated patches of habitat; (3) they require habitats that are diverse in structure and food supply year-around, which makes them largely dependent on suitable habitats; (4) they are an important resource for food, leisure activities, aesthetic value and symbolise a part of biodiversity in a concrete way; (5), as a point of practical relevance, there exist often longer time series of count and, especially, of harvest data. Examples of the use of game species as indicator species are ungulate species for forest management (Hanley, 1993; Liu et al., 2018) or food availability in the context of forestry (Jędrzejewska et al., 1997), ground-nesting birds for changes in agricultural use, especially also for the technical progress in agricultural production that has caused a monotonisation of habitat conditions (Klansek, 2002), or small and medium-sized carnivores for landscape structures (Ozoliņš and Pilāts, 1995). At the same time, different species can also provide indications on the interaction of habitat suitability and quality or weather conditions, such as ungulates (Jędrzejewska et al., 1997) and lagomorph species (Frylestam, 1980; Rödel and Dekker, 2012), or can serve as indicators for natural and management-induced interspecific competition and predation (Jędrzejewska et al., 1997; Ozoliņš and Pilāts, 1995). Thus, the results so far suggest that wildlife population dynamics are suitable indicators of landscape quality change. Despite all the advantages of game species as indicators, however, it must not be forgotten that game species are largely managed (or at least influenced) populations. Population dynamics thus reflect not only the environment but also, to some extent, management activities such as hunting per se.

Ideally, rather than using individual species as indicators, several species should be combined so that natural and artificial fluctuations in populations are less significant and long-term dynamics can be better depicted (Foster et al., 2002; Vierikko et al., 2010). For example, for the analysis of the ecological status and dynamics in forestry (Vierikko et al., 2010), the authors used a bundle of different 17 forest species. For the forest bird index and the farmland bird index of the European Union, 28 respectively 33 common bird species associated with forests or farmlands were combined (European Environment Agency, 2016; Gregory et al., 2005). In most studies to date, only individual species or specific groups of species have been used to analyse developments in specific socio-ecological systems. However, the extent to which a broad range of species that inhabit a wide variety of ecosystems provides an overall view of a region has only rarely been investigated (e.g., Foster et al., 2002) in New England. With this study, we would like to make a significant contribution to the closing of this knowledge gap. Using the example of South Tyrol (North Italy), a region in the centre of the European Alps, all 28 huntable game species are analysed simultaneously and examined with regard to their habitat preference and requirements. Such habitat guild analyses have been used to study the relationship between habitat and landscape features and different species groups (e.g., Benoit et al., 2021; Patterson et al., 2017). The habitat guilds studied are groups of alpine, forest and farmland species as well as species of settled areas, but they also comprise species that inhabit large landscapes with different habitats. Thus, these habitat guilds cover most of the landscapes and habitats found in South Tyrol.

Since real monitoring data based on count data are rarely available and, especially, not in long time series, harvest data are frequently used as indicators of population abundance. Various studies with ungulates (Bender and Spencer, 1999; Imperio et al., 2010; Jędrzejewska et al., 1997; Liu et al., 2018), carnivores (Ozoliņš and Pilāts, 1995), lagomorph species (Ferreira et al., 2010; Frylestam, 1980; Rödel and Dekker, 2012) and birds (Cattadori et al., 2003; Klansek, 2002; Martinoli et al., 2017; Ranta et al., 2008; Soininen et al., 2016) have shown that harvest data in long time series analyses are indeed suitable as estimates of population dynamics. Of course, attention is always advised when using harvest data as reference data for population abundance. The quality of harvest data for population abundance depends, among other aspects, on the species. For

ungulates, the correlation is usually quite good, and the harvest date show the pattern of population dynamics well (Bender and Spencer, 1999; Imperio et al., 2010; Jędrzejewska et al., 1997; Liu et al., 2018), but various studies have also shown that for smaller game species, this relationship is not always clear and that harvest data only partly reflect population dynamics (Ferreira et al., 2010; Frylestam, 1980; Ranta et al., 2008; Rödel and Dekker, 2012). This may be due to a lack of systematic procedures in data collection, changes in hunters' interest in certain game species and sporadic action due to different hunting systems (Ferreira et al., 2010; Imperio et al., 2010; Martinoli et al., 2017), but also because the census data are overdispersed as a result of an uneven count effort and count methodology among hunting districts (Foster et al., 2002; Martinoli et al., 2017). It is therefore important, for studies that use harvest statistics, to first obtain an overview of the quality of the data and to check whether the present hunting system can provide suitable data at all. In our study area, South Tyrol (Italy), harvest statistics have been collected centrally at the hunting ground level by always the same authority and according to a uniform standard since 1950. A considerable problem also for our study lies in the rare count data for the game species. High quality count data, which were collected according to a uniform standard (same counting method, season and spatial extent) and are therefore suitable for a correlation with the harvest statistics exist for 6 species.

In sum, we tested the following three hypotheses with the present study:

- Harvest statistics are suitable surrogates for population abundances under the South Tyrolean hunting system.
- Species with similar habitat requirements show comparable population dynamics in space and time. Thus, general dynamics of habitat guilds can be derived for the different habitats.
- Such guild dynamics can be used as a bioindicator to quantitatively and qualitatively describe changes in landscape quality for game species of an entire region.

2. Material and methods

2.1. Study area

Our study area is the Autonomous Province of South Tyrol in the Central Alps in northern Italy (Fig. 1), which covers an area of 7399 km². Sea level ranges from a minimum of about 194 m a.s.l. to a maximum of 3893 m a.s.l., and about 40 % of the entire area lies above 2000 m a.s.l. The region extends over two major geographical units, i.e., the Central Alps in the North and West and the Southern Limestone Alps in the Southeast of the study area.

South Tyrol is characterised by a mixture of semi-natural habitats and areas with strong anthropogenic influence (Fig. 1A). Overall, 48.4 % of the area is covered by forests, predominantly spruce and pine forests, with only 1 % being deciduous forest. Another 10.6 % of the area is intensively used as agricultural area, such as intensively fertilised hay meadows, arable areas and permanent crop areas used for apple and wine production (Hinojosa et al., 2019); 11.9 % is covered by low-stocked alpine summer pastures, non-fertilised mountain hay meadows and traditionally used agroforestry systems (e.g., larch meadows, fruit and chestnut orchards). Furthermore, 26.5 % of the region is made up of alpine and nival habitats such as grasslands, rocky areas, screen slopes and glaciers, 0.3 % is wetlands and water bodies, and 2.3 % is urban and rural settlement areas. This land use/land cover (LULC) distribution is a snapshot and has changed significantly over the past decades: The pressure on land is most pronounced on the valley floors in the colline belt, where the majority of inhabitants live and the most intensive forms of agriculture are found. Accordingly, this zone has changed massively over the past 150 years (see Fig. 1B), and 70 years ago, the climatically favourable zone of the colline belt was still dominated by a mosaic of fertilised grassland and arable land. Today, this area is dominated by fruit and wine monocultures. In the agriculturally used montane zone, on the other hand, arable land has largely been converted into intensively fertilised hay meadows. Less

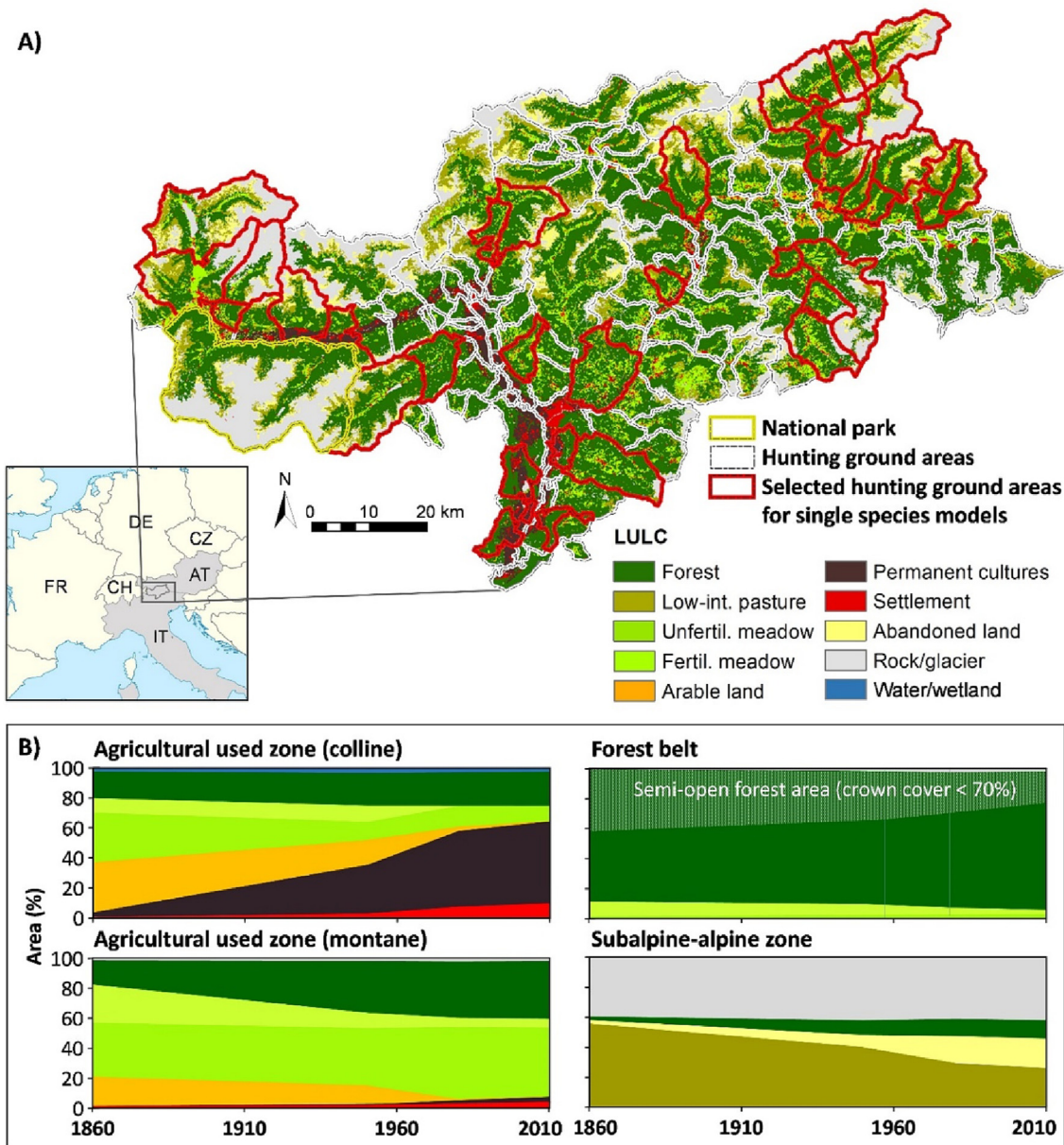


Fig. 1. A) Location of the study area land use/land cover (LULC) distribution after Rüdiger et al. (2015) and hunting ground areas. B) LULC changes since 1860 for the most important landscapes (changed after Tasser et al., 2009).

favourable agricultural areas in the subalpine and alpine belts are increasingly being abandoned, especially on the less accessible and steep slopes (Tasser et al., 2009). As a result, the forest area has expanded in many areas, with a massive densification of the forests. Whilst 150 years ago, more than half of the forests were still semi-open (crown cover <70 %), this proportion has decreased to 20 %.

2.2. Hunting situation

Hunting games have a long tradition in South Tyrol and a varied history (Erhard, 2006). Until the 19th century, hunting was reserved to nobility and clergy. In 1848, a break in the hunting law came into force in the Habsburger Empire (of which South Tyrol was part of). In large parts of the Empire, there were uprisings because of the 'hunger winter' of 1847/48. The high lordly hunting rights were abolished, and the rural population could now hunt for food. This led to a strong decline in various game species populations. A whole range of game species was exterminated in Tyrol, including brown bear, wolf, lynx, ibex, but also beaver, otter and

bearded vulture (Unterthurner et al., 2019). In 1875, the Austrian imperial hunting act was established to regulate hunting within the empire borders. These regulations applied to South Tyrol until the First World War (1914–1918). After the annexation of South Tyrol by Italy, Italian hunting law was applied in South Tyrol. In 1950, the province of South Tyrol became autonomous and introduced various regulations over time, which then merged into its own hunting law in 1987, which are still valid today. In shooting plans, closed seasons and harvest limits were set for single game species, and hunting with snares was prohibited. These administrative hunting restrictions did not remain static over time, but were temporally and spatially adjusted to population trends according to the counts or observations. Hence, most changes in harvest plans reflect population changes. To compensate for short term temporal variations, we did not use the hunting statistics of individual years, but the average over 10 years. Hunters had to register, and over time, compulsory hunter training and licensed hunting followed. Administratively, South Tyrol was split into 145 hunting grounds (Fig. 1A), harvest statistics were collected on a yearly basis for each hunting ground, and the hunting administration

was delegated to the South Tyrolean hunting association (for more information, see [Unterthurner et al., 2019](#)). Therefore, harvest statistics were available from 1875 to 1895 and then again yearly for 1953 onwards for every hunting ground.

2.3. Characterisation of harvested species

There are harvest statistics for a total of 28 species in the study area (Table 1). Several of these species have similar habitat preferences and requirements and have therefore been grouped together in habitat guilds:

A) **Ungulates:** Wild ungulates show a high ability to tolerate anthropogenic land use and habitat modifications, especially wild boar, roe deer and red deer ([Linnell et al., 2020](#)). These species prefer large and highly diverse forests but can survive well in a mosaic of open cultivated areas and forests ([Bradshaw et al., 2003](#)). The situation is somewhat different with the chamois. For a long time, it was considered a species that occurs mainly in subalpine and alpine terrain. In the past decades, however, it has increasingly dispersed to forest sites at lower elevations ([Fankhauser and Enggist, 2004](#); [Suchant et al., 2003](#)). Negative impacts on habitat quality result from various infrastructures (e.g., roads, fences, energy production plants, tourist resorts) and forms of human disturbances (e.g., hiking, skiing) ([Beyer et al., 2016](#); [Linnell et al., 2020](#)).

B) **Mesocarnivores:** These mammalian predators require large, forest-dominated habitats and are sensitive to habitat loss and fragmentation ([Červinka et al., 2014](#)). In contrast to large specialist carnivore species, some mesocarnivores (e.g., stone marten, red fox and Eurasian badger) have largely expanded in the last centuries into the open rural and urban environments due to the higher food availability ([Bateman and Fleming, 2012](#)). The pine marten, on the other hand, avoids open spaces to a large extent but likes to spend time at the edge of forests and hedges. Its presence is therefore not restricted to forests, but it is, to a certain extent, dependent on the presence of trees in the surrounding area ([Pereboom et al., 2008](#)).

C) **Small alpine games:** All three species studied (mountain hare, marmot, rock ptarmigan) occur preferentially at subalpine and alpine elevations. They prefer open, forest-free habitats with various protection and feeding possibilities ([Galluzzi et al., 2017](#); [Rehnus et al., 2018](#); [Schweiger et al., 2012](#)). The habitats must contain a mix of dwarf shrub, rock and grassland cover. However, the snow hare can also be found in forests up to about 1300 m ([Rehnus, 2013](#)), but always in rough terrain with abundant grasslands as this landscape configuration provides a diverse mosaic of food and protection in close distance ([Rehnus et al., 2018](#)).

D) **Small forest games:** This group contains only bird species that have their habitat exclusively or largely in forests. The capercaillie, the hazel grouse, prefers intact mature and old-growth forests as well as forest management practices that emulate natural disturbance processes ([Mikoláš et al., 2015](#); [Schäublin and Bollmann, 2011](#)). Woodcocks are more abundant in woods containing a more heterogeneous mix of woodland habitat types and in woods further from urban areas ([Hewar et al., 2018](#)), whereas the black grouse prefers the subalpine zone with semi-open forests ([Schweiger et al., 2012](#)). The Eurasian jay typically chooses forests during the breeding season but can then also be found occasionally on agricultural land or even near settlements ([Gianpasquale and Alberto, 2019](#)).

E) **Small farmland games:** In addition to the typical farmland birds (grey partridge, common quail, common pheasant, rock partridge, common wood pigeon; ([Gregory et al., 2005](#)), the brown hare also belongs to this group. The ideal habitat of these species consists of a mosaic of different habitat types (e.g., arable land and horticultures, grasslands) with boundary and linear features such as hedgerows and, ideally, with standing open waters. Such areas are home to a high biodiversity, offer sufficient feeding and nesting facilities and provide protection areas. Monotonous and intensively used agricultural landscapes are less suitable as habitats for most farmland species ([Scholefield et al., 2011](#)).

Synanthropic games: This group includes five game bird species (Eurasian magpie, carrion crow, common blackbird, song thrush, fieldfare) which respond positively to urban development ([Rodewald and Gehrt, 2014](#)). These species are generalists, omnivorous and flexible and

accordingly occur in a wide variety of habitats: forests, agricultural landscapes and settlement areas. However, the highest densities are found in orchards, vineyards and urban areas, depending on the amount of fodder they feed on.

Waterfowls: In South Tyrol, four waterfowl species are huntable: mallard, garganey, common teal and the Eurasian coot. Until 1985, however, no distinction was made among the duck species, and only a total sum was documented. Since mallard ducks have made up 96 %–98 % of the individuals harvested in recent decades, it can be assumed that they were the main species harvested before.

2.4. Methods

This study analysed species population and habitat guild dynamics over the last 150 years in South Tyrol and used them as bioindicators for habitat and landscape (as a heterogeneous geographic area characterised by diverse interacting habitats or ecosystems) quality changes for game species in the region. Several methodological approaches were used ([Fig. 2](#)). Our first hypothesis was that harvest statistics adequately reflect the population dynamics of game species. To test this, we searched for count and monitoring data for as many species as possible and correlated these population data with the harvest data. The second hypothesis states that species with similar habitat requirements (so-called habitat guilds) show comparable population dynamics in space and time. This second hypothesis was tested by recording long-term harvest series for all game species in all 145 hunting areas of South Tyrol by comparing the single population dynamics. Finally, based on the dynamics of habitat guilds, trends in landscape quality of habitats or, in combination with habitat guilds that use larger and more habitats, landscape quality changes of the entire landscape were modelled. We defined landscape quality in terms of game species by the number of harvested habitat guild individuals. This means that the more productive populations are supported, landscape quality is higher because the habitat conditions are better in this landscape. Compared to traditional landscape indicators such as landscape metrics, a significant advantage in using animal populations as indicators of landscape quality is that changes in habitat quality are also taken into account. For example, the quality of many agricultural habitats has changed massively over time, as hay meadows are fertilised and cut more frequently nowadays, and the use and amount of different pesticides in arable farming, orchards and vineyards has increased.

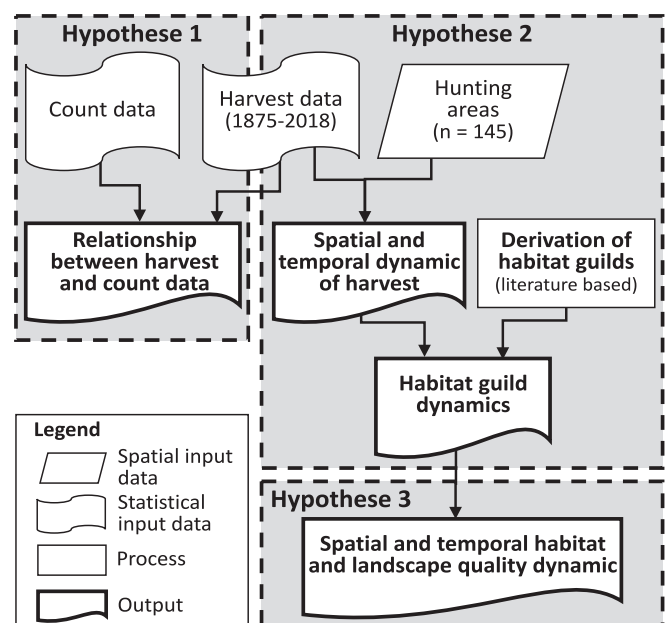


Fig. 2. Conceptual design used in this study.

2.4.1. Harvest data curation

The dynamics of species populations since 1890 was reconstructed by using harvest data. Historical data from the end of the 19th century were derived from statistical reports of the *Handels- und Gewerbekammer Bozen* (1894) and the statistical yearbooks of the Ministry of Agriculture (K.K. Ackerbau-Ministerium, 1896, 1893, 1881, 1876). From 1953 onwards, the South Tyrolean hunting association provided yearly harvest statistics for the 28 game species on the hunting ground level (see Appendix 1, Fig. 1). Since the introduction of the hunting system in South Tyrol in 1950, the hunting system has changed only insignificantly and the survey standards have also remained uniform over the period. In addition, strict attention was paid throughout the time to ensure that all harvested animals were properly reported by hunters (Erhard, 2006). While in the 19th century there were only a few hundred professional hunters, over time the number of amateur hunters has increased to an average of 6300 (Unterthurner et al., 2019). The spatial resolution of harvest data is defined by the 147 hunting grounds, which covers an area from 2.3 km² to 302.4 km² with a mean of 48.3 km². All these conditions ensure a high quality of the harvest data in South Tyrol.

Although hunting statistics from the 19th century were also collected at hunting ground level, the original data were lost in the course of time, so that today those statistics are only available at district level. The conversion of the data to the level of hunting grounds was performed by the ratio of the mean harvest data from the period 1953–1964. This conversion is subject to some uncertainty.

2.4.2. Relationship between harvest and count data

Harvest statistics do not automatically reflect population dynamics. Our hypothesis, however, was that the South Tyrolean hunting system is suitable to reflect population dynamics via long-term harvest data. To test this, we searched for independent count data for as many game species as possible and correlated these population size data with the harvest data (Fig. 3). Species with available independent population count data were red deer, roe deer, chamois, black grouse, rock ptarmigan and rock partridge. For all species except roe deer, these data were obtained from regional monitoring conducted over the past 20 years by the state authority (Hunting and Fishing Office) in cooperation with the hunting association and local hunters in so-called population units (larger, contiguous species-specific habitat complexes). Rock partridge population surveys were conducted in April–May on several fixed sample plots using decoys (for methodological procedure see also (Schemnitz et al., 2012)). For the Rock Ptarmigan, additional counts were carried out on the designated sample plots in summer using pointing dogs, which is an established method for the Alpine region (see Novoa et al., 2019). Black grouse were counted in spring at all known courtship sites in South Tyrol. All cocks seen and/or heard were counted, and individuals in spatially contiguous courtship

sites were usually counted on the same day to minimize the risk of double counting (Dumont et al., 2019). Chamois were monitored throughout South Tyrol via block counts, counting individuals from fixed vantage points or along trails using optical instruments (Obber et al., 2022). The survey area was divided into smaller and manageable “blocks”. Double counts were avoided by counting on the same day, with animals migrating between blocks corrected out. The red deer population in South Tyrol is surveyed in open, forest-free areas via spring spotlight surveys (Corlatti et al., 2016). This method is a proven field methods to collect data on abundance relies on the use of spotlights during night roadside counts. In the case of roe deer, a species that is highly difficult to monitor, the data come from a long-term survey by one single hunter who had always used the same method to count the individuals in his home hunting ground each year. These counted population size data were correlated with the harvest data.

2.4.3. Species population dynamics

A mean number of harvested individuals per hunting area (mostly a 10-year mean) was calculated for the time periods 1875–1895, 1953–1964, 1965–1974, 1975–1984, 1985–1994, 1995–2004 and 2005–2014. This time resolution allowed for the identification of significant long-term population dynamics while also filtering out much of the short-term variation that is observed in yearly harvest data. Harvest time series differ in length among species. Some species, such as red deer, roe deer, chamois, capercaillie and black grouse, have been recorded already since the end of the 19th century, whereas different thrush species, for example, were not recorded at the species level (only the total number of harvested small birds was determined). For these species, data are only available from the 1980s onwards. Furthermore, certain species, such as the capercaillie, were closed for hunting due to critical population dynamics, resulting in a changing species protection status. Detailed information about the time series of the single species can be found in Appendix 1.

2.4.4. Derivation of the dynamics of landscape quality

Changes in habitat and landscape quality are derived from the habitat guilds. This approach is based on the hypothesis that the highest mean harvest numbers occurs at a time that is most ideal for the habitat guilds in terms of the overall life conditions. Johnson (2007) and Fukasawa et al. (2020) showed that for wild species, population abundance is not always the best indicator for determining habitat quality. In their opinion, several demographic indicators would have to be used together, but this is rarely done due to the difficulty of collecting such data, and certainly not available from past times. Thus, results based solely on abundances (or in our case on harvest data), even if they result from different species of a habitat guild, should be interpreted with caution. In addition, hunters' interest in game species can also severely disrupt the relationship. To account for this effect,

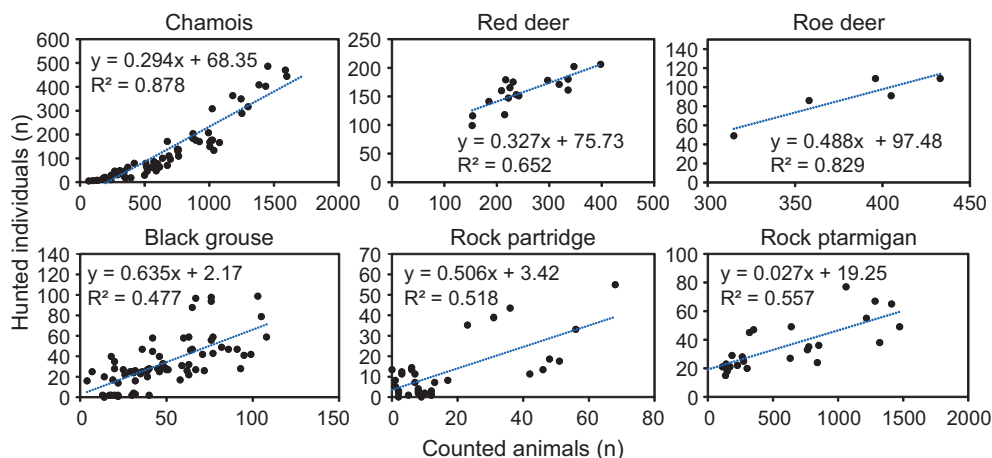


Fig. 3. Linear regression between harvest data and count data of six game species. All correlations are significant ($p < 0.05$).

we surveyed in interviews with hunting managers changes in the interests and considered these as confounding factor in the evaluation.

The assignment of the species to the habitat guilds was based on literature data to the species-specific habitats (see Appendix 2). We calculated habitat quality trends per single hunting area for the alpine-nival habitat,

the forest habitat, the farmland habitat (separated according to colline and montane elevation levels), for the settlement and the water habitats. In addition, we derived information about large-scale changes in landscape quality from the guild trends of ungulates and mesocarnivores. Landscape quality was calculated using several steps: 1) calculation of mean harvest

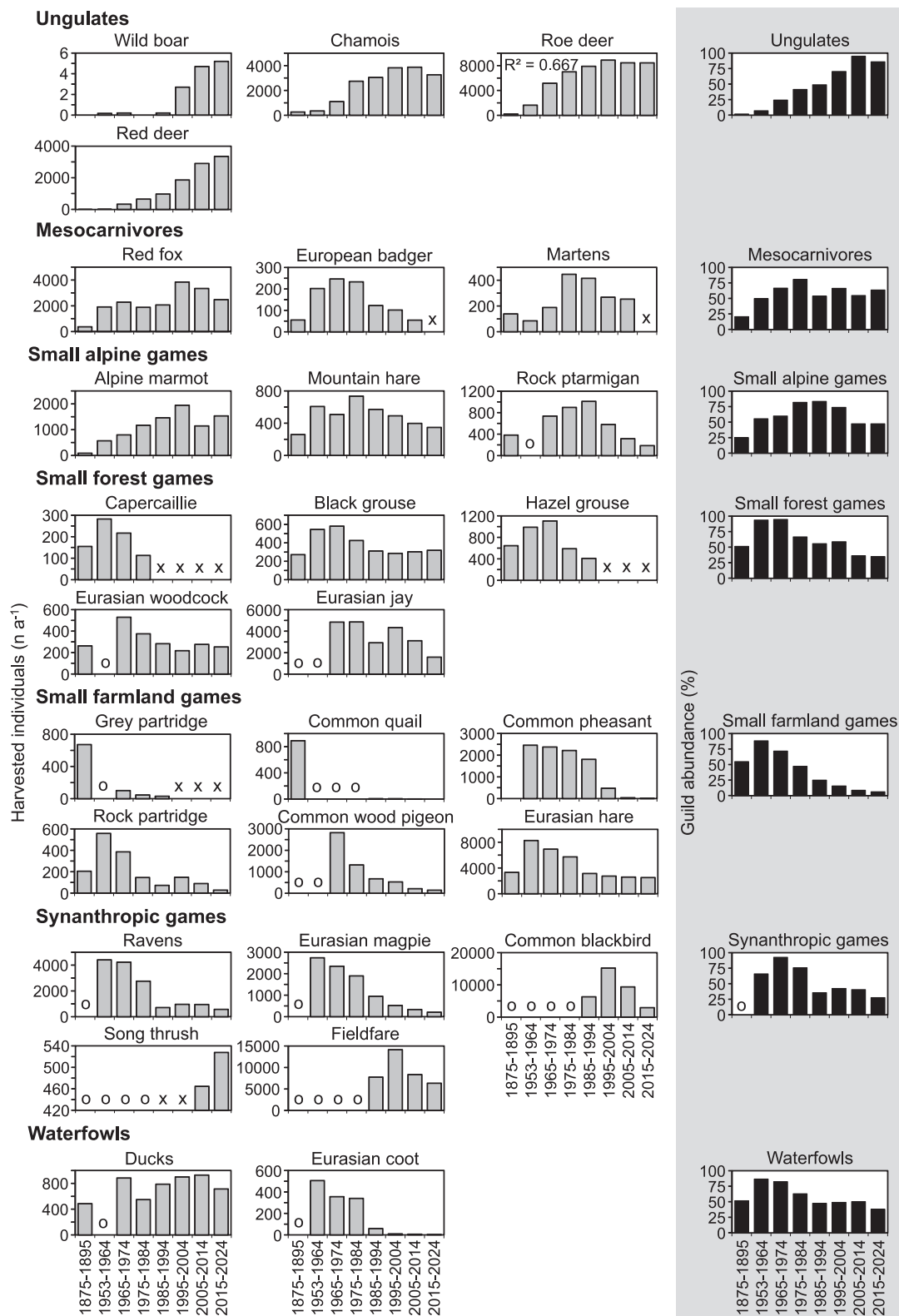


Fig. 4. Development of harvested individuals per game species in South Tyrol from the end of the 19th century, summarised according to habitat preferences, as well as the mean habitat guild harvest (mean of all max normalized species values within the habitat guilds). o = no data, x = protected.

number per species ($\text{n ha}^{-1} \text{y}^{-1}$) based on available species-specific habitats (Appendix 2) in the hunting ground and per 10-year time period; 2) calculation of mean harvest number ($\text{n ha}^{-1} \text{y}^{-1}$) per time period across all hunting grounds, i.e., a mean harvest for South Tyrol; 3) calculation of the relative species-specific mean harvest per time period for South Tyrol, normalising over the maximum occurred mean harvest over the time periods (max-normalisation); 4) calculation of the mean guild harvest per time period over the normalized harvest of all species within the habitat guild (see result in Fig. 4). For the spatial representation of the current landscape quality (see Fig. 5), we calculated the normalized harvest (expressed in % of the maximum) per game species and hunting ground for the period 2005–2014 (most recent, complete period). Then, for each hunting ground and LULC type the mean harvest was calculated from the harvest number of all game species occurring in this LULC type (see Appendix 2). The spatially explicit representation was calculated using the current LULC map (resolution: $5 \times 5 \text{ m}$, see Fig. 1).

To analyse the change in landscape quality over time, the corresponding max-normalized harvest number for each 10-year period was calculated for each hunting area and then averaged across all hunting areas. The mean landscape quality was calculated on all species occurring in the individual LULC types as well as only on the LULC-specific specialists (see chapter 2.3). For more details see Appendix 5 (original calculation file).

2.4.5. Statistical analysis

To describe the species, guild and landscape trends, descriptive statistics was used and to investigate the relations between the harvest data and independent count data a linear regression was applied. All calculations were conducted in SPSS software version 28.

3. Results

3.1. Relationship between harvest data and independent count data

Independent population count data of six species were correlated with harvest data. Harvest data and count data depicted strong linear relationships, with coefficients of determination (R^2) ranging from 0.47 to 0.88 for the analysed species (Fig. 3). Especially high coefficients were identified

for chamois ($R^2 = 0.88$) and roe deer ($R^2 = 0.83$). The lowest level of coefficient of determination was detected for the black grouse, with an R^2 of 0.47. Generally, relationships were higher for ungulates than for bird species. The results from the six species show that the harvest data are suitable as measures for population dynamics, and therefore, for further steps, an existing correlation between population abundance and harvest data for all animal species was assumed.

3.2. Habitat guild dynamics

Based on the 10-year means of the harvest data, the population dynamics was traced for all 28 game species. Most game species showed clear temporal and spatial dynamics (Fig. 4; Appendix 3): For the ungulates, an increasing harvest could be observed. This also applied to the wild boar in recent decades, which was mainly exterminated in South Tyrol before 1995. For mesocarnivores, in initial increase in populations was observed, followed by a moderate decrease after 1970/80. The same development was observed for small alpine species: After an initial increase, there was a decrease after 1980/1990. A decreasing dynamic just after the middle of the 20th century could be observed for small forest games, with capercaillie and hazel grouse being banned from hunting from 1984 and 1986, respectively, due to the sharp decline in numbers. The numbers of harvested black grouse, woodcock and Eurasian jays have stabilised at a low level. A strong decline in the harvest was marked for small farmland games. The grey partridge, the common quail and the common pheasant largely disappeared from the South Tyrolean landscape. The populations of the synanthropic game species (ravens and Eurasian magpie) have also been declining sharply since 1960/1970. The common blackbird and the fieldfare showed a decrease from 1990 onwards, whereas the data series for the song thrush is too short to make a statement. Finally, the waterfowls showed different dynamics: whilst the number of Eurasian coots decreased significantly, that of ducks remained constant.

Starting from the individual species, habitat guild dynamics could be derived in different habitats and landscapes over the last 150 years (Fig. 4). Initially, the harvest of all habitat guilds increased everywhere, followed by a strong decrease in farmland and synanthropic species. In the forest belt and in the subalpine/alpine belt, the decrease in typical species was considerably lower.

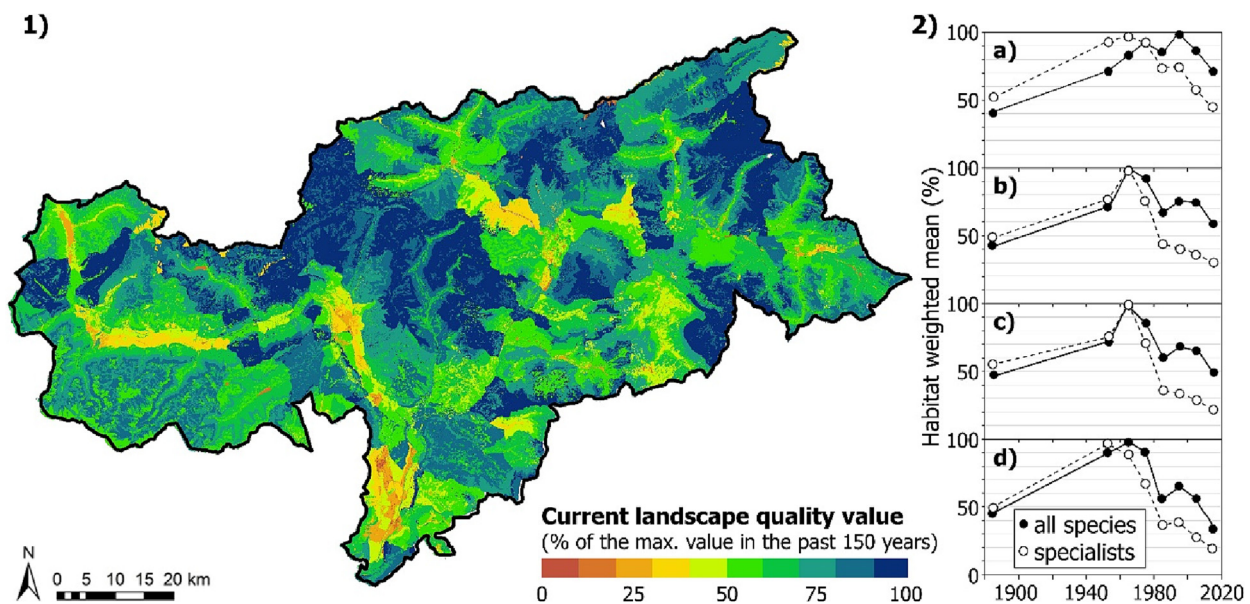


Fig. 5. Map of the current landscape quality situation (2005–2014) with regard to the game species. 1) The map reflects the current quality value as a % of the maximum value in the single hunting areas within the last 150 years. 2) Changes in landscape quality based on historical harvest data within the regions: a) subalpine-alpine grassland, b) forest belt, c) agriculturally used montane belt, d) agriculturally used colline belt. The two lines indicate the changes considering all game species living there respectively using only on the habitat specialists.

3.3. Habitat and landscape quality changes

Assuming that highest harvest occurred at the time with the best habitat conditions for the individual species, the qualitative change of the landscape was modelled for all single hunting areas in South Tyrol (Fig. 5). The result showed that the current situation was, in a large area, not optimal, especially at lower elevations. Many habitats that occurred there have deteriorated qualitatively for the game species. On intensively used agricultural areas, the values were between 30 % and 40 % of the maximum value of the past 150 years (see Appendix 4). Particularly affected by a decline in quality were the broad valley bottoms with orchards and vineyards. However, the quality value had more than halved, even in the typical intensive grassland areas. The quality values of many forest habitats at lower elevations ranged from 65 % (hardwood forest) to 76 % (oak forest) of the maximum value. Forests at higher elevations, on the other hand, showed values between 73 % (green alder and mountain pine forests) and 90 % (Krummholz) of much higher values. Their conditions thus largely corresponded to the maximum value of the past 150 years. With values between 85 % and 92 %, many alpine and nival habitats were also in a very good condition.

Considering only the habitat specialists (i.e., excluding the ungulates and the carnivores), the decrease in landscape quality value was much more pronounced (see Fig. 5). Currently, in the colline, intensively used habitats were mostly even, in a range of 30 %–60 % of the maximal value, whereas in the montane region between 800 and 1400 m a.s.l., the values were in a range of 60 %–70 % of the optimal condition. In higher regions (< 1400 m a.s.l.), on the other hand, the current situation of the habitats largely corresponded to the optimal state with regard to the typical game species, with values between 80 % and 100 %.

4. Discussion

Our study is based on two strongly methodological hypotheses and one more substantive hypothesis. First, that it is possible to assess population dynamics based on harvest data, and second, that species with similar habitat requirements show comparable population dynamics in space and time and thus can be grouped into habitat guilds. The third hypothesis states that guild dynamics can be used as a bioindicator to describe changes in landscape quality. Previous studies have already shown that the use of population dynamics of wildlife species is an ideal measure for analysing habitat and landscape changes as wildlife species attribute to changes in the availability of key resources, such as habitat structures, food and nest sites (Davies et al., 2010; Hanley, 1993; Klansek, 2002). Changes in key resources in cultural landscapes are highly connected with changes in management and, thus, with the socio-economic developments (Foster et al., 2002; Scholefield et al., 2011; Vierikko et al., 2010). Conversely, the changes in species harvest or, even better, in habitat guild harvest can provide relevant information about positive or negative dynamics in management and landscape quality, even if they are not immediately visible.

4.1. Methodological considerations

The methodological approach used in this study provides evidence for the usefulness of game data as a measure of population dynamics (Hypothesis 1) and as indicators for tracking landscape quality development over a long period of time (Hypothesis 2). Due to the high number of game species with different habitat requirements, it was possible to make statements about specific habitats as well as about landscapes. In our view, habitat specialists are particularly valuable for the interpretation of landscape development. Through them, for example, the consequences of the intensification of agricultural use or the consequences of the densification of forests can be identified. The development of generalists, such as ungulates, does not allow such specific statements, especially because some species have been largely exterminated in the 19th century (see red deer and wild boar), and therefore, the population in the study area had to re-establish itself slowly. The chamois and the roe deer already reached their distribution

optimum in the 1990s, and it can therefore be assumed that the population development since then has mainly been caused by habitat suitability. Habitat quality seems to be largely constant or slightly deteriorated. These examples show that shooting numbers can make a valuable contribution to the interpretation of landscape development.

However, our methodological approach has some limitations. First, our study was designed to capture the population dynamics based on harvest data. This relationship has already been tested for many game species and has proven to be effective for many species (Cattadori et al., 2003; Imperio et al., 2010; Jędrzejewska et al., 1997; Martinoli et al., 2017; Ozoliņš and Pilāts, 1995; Soininen et al., 2016). However, in hares and rabbits, but also in grouse species, there may well be deviations in the derived dynamics based on the counting and harvest data (Ferreira et al., 2010; Ranta et al., 2008). In addition, the harvest data cannot be converted equally into absolute population densities in all hunting grounds since they largely depend on the hunting management. Some authors (Martinoli et al., 2017) have demonstrated that adding the game management authority as a model covariate can lead to more robust trend estimates, indicating a significant effect of different hunting policies and hunting efforts. A validation of the correlation between harvest data and count data, as we have done for six game species, is therefore necessary and can prevent misleading estimates of abundance indices (see also Imperio et al., 2010).

Another problem with the use of harvest data can be that the interest in certain game species may have changed over the past decades. For example, a survey of game wardens in South Tyrol shows that hunters' interest has shifted from small game to ungulates (Unterthurner et al., 2019). Certain species that are difficult to hunt without pointing dogs (e.g., woodcock, ptarmigan) are hunted less today than in the past because of the small number of dog owners among hunters. In the case of South Tyrol, however, this statement is contradicted by the constant shooting of ducks and marmots. For these species at least, the interest today is approximately the same as it was 50 years ago. The number of harvests is also influenced by targeted protection measures. For example, certain species have been placed under partial or complete protection several times over the course of many years. In many cases, this was the result of a sharp decline in population abundances at national or regional level. Examples include the capercaillie, the hazel grouse, the pine and the stone marten, as well as the European badger. Some of these populations have been able to stabilise over the past 20 years, as evidenced by the number of carcasses found, which were examined by the State Veterinary Service. This is specifically the case for the badger and the martens in South Tyrol (www.provinz.bz.it/landwirtschaft). Thus, the harvest data for these two species at least could underestimate the actual population abundance. And finally, there is a specificity regarding the pheasant. This species is not native to South Tyrol, but it has already been released for hunting purposes in the middle of the 20th century, and the population was artificially kept at a high level by annual additional releases (Unterthurner et al., 2019). From 1997 onwards, a ban on release into the wild was imposed, and the population quickly collapsed. Today, only isolated individuals can be observed in South Tyrol (Unterholzner, 2017).

4.2. Landscape quality changes in South Tyrol

Regarding the third hypothesis, our results prove that habitat guilds indeed indicate different developments in the quality of the main habitats and landscapes in South Tyrol.

The alpine-nival habitats improved qualitatively until the 1970/80s of the last century when considering the specialists found there; since the 1980s, especially for the populations of mountain hare and rock ptarmigan, the habitat has been deteriorating significantly, whereas the populations of the marmot have been classified in the last management plan (2022–2026) as stable. The mountain hare and the rock ptarmigan are particularly sensitive to reforestation and afforestation of open grassland areas as a consequence of abandonment of summer pastures (Furrer et al., 2016; Hulbert and Iason, 1996), in addition to climate change (Imperio et al., 2013;

Rehnus et al., 2018) and disturbances by outdoor activities and pastoralism (Furrer et al., 2016a; Rehnus et al., 2014). On the other hand, marmots prefer, in contrast to the other two species, anthropogenic sites (meadows, summer pastures), where their reproduction rate was highest (Ramousse et al., 2009). Thus, marmots are competing with farmers for space, and especially on areas with increased pastoralism intensity, the marmot is increasingly in competition with livestock (Poudel et al., 2015), which explains the slight decrease in population densities. In summary, the alpine habitat in South Tyrol has clearly lost in habitat quality due to a decrease in agriculturally used areas and the accompanying reforestation, as well as the partly too intensive use of remaining areas by agriculture and tourism.

Forest habitats: The numbers of the typical forest grouse species (capercaillie, black grouse, hazel grouse) and the Eurasian woodcock have significantly increased until the middle of the 20th century and then slowly decreased until the end of the 20th century, when they stabilised at a low level. All these species have profited from the rigorous hunting management, but they have then suffered because of developments in forestry and agricultural activities. In South Tyrol, as in many Alpine countries from 1950, the forests in the interior have become much denser due to the marginalisation of forestry and the separation of forest and pasture (Seidl et al., 2019; Tasser et al., 2009). As a result, the understory vegetation has been significantly reduced, whereby the breeding success and the survival of these species depend, above all, on the presence of insects (in the chick phase), bilberry and cowberry (Baines et al., 2004; Summers et al., 2004). Thus, the quality of the forest habitats regarding the forest game species has slowly deteriorated due to increasing densification.

Farmland habitats: As also shown by other studies from Europe (Kamieniarz et al., 2013; Klansek, 2002; Scholefield et al., 2011), the farmland game species in South Tyrol have declined particularly sharply, and some of them have even disappeared from the landscape. This negative dynamic was also triggered in South Tyrol, primarily by land-use change and an intensification in agriculture. Arable land is being cultivated; many structures (e.g., hedges, stone walls) have been removed, and wetlands have been drained (Tasser et al., 2009). Traditional orchard meadows have given way to modern orchards, where highly effective pesticides are increasingly being used, and hail nets are installed. All these developments are leading to a reduced supply of key resources, such as food and nest sites, for many of these species (Klansek, 2002; Scholefield et al., 2011). In South Tyrol, therefore, the agricultural area as habitat for these farmland species has qualitatively deteriorated from the middle of the last century.

Settlement areas: According to a previous study (Guetté et al., 2017), synanthropic species are mainly granivorous, aerial insectivorous, ground foragers, cavity nesters, sedentary birds, social birds, species with a wide latitudinal and elevation tolerance as well as exotic species. In the cities of Central Europe, urbanization is leading to an increase in densities and biomass but maybe also to a reduction in species richness (Shochat et al., 2015; Tryjanowski et al., 2017). In South Tyrol, however, this positive trend in connection with the increasing urban sprawl is not evident. This is probably related to the fact that settlement areas in South Tyrol are generally small, and therefore, these species are massively influenced also by the surrounding agricultural areas (see also Clergeau et al., 2002). Accordingly, it is not surprising that their populations are declining in South Tyrol, just as those of the farmland species.

Water habitats: The dynamics in the two waterfowl species do not allow the assumption of a consistent trend in water habitat quality. However, some indications emerge: Areas in which open land is closely interlinked with water bodies are suitable duck habitats (Davies et al., 2010; Unterholzner, 2017). Although ducks seek their food both on land and in the water, they nest on land. According to our results, these species benefit from highly structured agrarian habitats but also from an intensification of grassland farming and the associated increase in the frequency of cutting, which makes it easier to find food on land (see also Schmieder et al., 2006). The development of the Eurasian coot is clearly different. The population of this species has been decreasing significantly since 1980, which has also been observed in other regions of Europe (Lehikoinen et al., 2016; Musilová et al., 2018). Coot populations are sensitive to food

resources, especially submerged macrophytes and aquatic insects, and are in competition with fish (Maceda-Veiga et al., 2017). In sum, the population trends of ducks and Eurasian coot indicate that habitat quantity in South Tyrol has not significantly changed, but the quality of existing wetlands (especially of eutrophic wetlands) may have decreased. This would be an alarming sign since the last century, more than half of the wetlands have been lost worldwide because of anthropogenic actions (Davidson, 2014).

Finally, ungulates and mesocarnivores still allow for conclusions about the overall development of the landscape quality beyond single habitats. In our study, all ungulate species (red deer, roe deer, chamois, wild boar) showed growing populations. Roe deer and chamois numbers have stabilised at high population levels in recent decades; the other two species are still in the process of dispersal in South Tyrol. The species have their habitat in the transition between forest and open land. An exception is possibly the chamois, which prefers open habitats in the subalpine and alpine belt, where more digestible food can be obtained, but this species also uses forest as habitat during 22 %–48 % of its lifespan (Fankhauser and Enggist, 2004; Garcia-Gonzalez and Cuartas, 1996). These species were all severely decimated or, in the case of red deer and wild boar, exterminated in the 19th century by the intense human hunting pressure in South Tyrol (Erhard, 2006). As our study shows, these species could primarily recover through regulated hunting management. But, as also shown by other authors (Foster et al., 2002; Liu et al., 2018; Vierikko et al., 2010), this positive development in a mainly cultivated landscape depends also on the expansion of forests as a result of the abandonment of agriculturally unfavourable sites and their qualitative improvement (Hinojosa et al., 2019; Tasser et al., 2009). Like the ungulates, the mesocarnivores also benefit from a reduction in hunting pressure and a large and diverse landscape of forests and agricultural land. The three dominant species in the area (red fox, European badger, stone marten) feed in varied ways. Besides insects, worms, small mammals, birds, wild fruit and carrion, they also benefit from cultivated fruit (Balestrieri et al., 2011; Prigioni et al., 2008). Thus, the development of land use in South Tyrol does not directly explain the decreasing population of these species since the 1980ies. However, if we look at agricultural land management, it becomes clear that both grassland as well as fruit and wine growing have been significantly intensified and that arable land has largely disappeared. Today, grassland areas are fertilised more intensively and cut more frequently, and many of the former extensive pastures have been abandoned or changed into hay meadows (Tasser et al., 2009). Plant as well as faunistic diversity, and thus the food supply, largely decreased (Hilpold et al., 2018a; Niedrist et al., 2009). These results underline that the overall landscape in South Tyrol has improved for these habitat guilds due to forest expansion and more extensive forest management in the otherwise intensively used landscape. For the carnivores, however, food shortage due to intensive agriculture has an increasingly negative effect.

4.3. Implications for landscape management and planning

The results of our study indicate the importance of habitat protection measures to conserve and promote game species populations. South Tyrol makes no exception from the Europe-wide trend of decreasing biodiversity especially in agricultural areas (Scholefield et al., 2011). Many species associated with agricultural landscapes are currently on the Red List or have disappeared regionally. Measures required by the EU to green agriculture and to protect species (Evans, 2012) have so far been insufficiently implemented in South Tyrol. Environmental measures, such as the designation of 5 % of the total agricultural area as high-diversity landscape features as compensatory areas and the promotion of organic farming, have little effect in its current form of implementation. This is the only way to explain why the situation for farmland species has not improved in South Tyrol, but also in other European regions (European Commission, Directorate General Environment, 2021). The EU has therefore intensified its efforts to counteract species extinction in the agricultural landscape with the Green Deal and the EU Biodiversity Strategy 2030 (European Commission, 2020; Hermoso

et al., 2022). The next few years will now show whether or not the implementation of the specified measures will have an effect in farmland.

The situation is better for forest and alpine habitat associated species: habitat quality and quantity for these species has decreased mainly due to LULC changes as a consequence of socio-economic development in agriculture. Species that are predominantly promoted by extensively used habitats, such as alpine summer pastures and agro-forestry systems (larch meadows, forest pastures), have lost habitat as a result (Hilpold et al., 2018b; Mayer and Huovinen, 2007). Society can actively counteract this by financially supporting the traditional use of such areas (Marini et al., 2011; Schulz, 2015). The example of South Tyrol, where a large number of such compensation payments are already provided for, shows that they are not always crowned with success (Hinojosa et al., 2019). As is the case everywhere in the Alpine region, low-input farming systems have declined significantly in South Tyrol (Tappeiner et al., 2008). Compensation payments do not seem to be able to sufficiently compensate for the additional labour costs and reduced product prices (Hinojosa et al., 2019) or there is simply a lack of human resources (Roeder et al., 2010). More emphasis should therefore be placed in future on personnel support for shepherding, forest maintenance and management of extensive land-use types.

Finally, our results reveal another development that will intensify in the future: the consequences of climate change. There is already a clear shift of forests to higher elevations (Körner, 2012; Tasser et al., 2017). This automatically reduces the natural habitat for alpine game species, which will lead to a slow disappearance in regions with a low proportion of alpine and nival vegetation belts (Chamberlain et al., 2013). As an adaptive measure, humans can only support the active use of summer pastures and artificially counteract the prevention of large-scale reforestation through targeted payments.

5. Conclusions

Based on our results, the following conclusions can be drawn for the development of landscape quality for game species:

- In South Tyrol, harvest data are suitable for deriving population dynamics of huntable species.
- The population dynamics of species with similar habitat requirements have developed similarly, and guild dynamics can be derived.
- Our results clearly underpin that in the future, the priority must be to improve the quality of farmland. A reduction of the intensity of use, a spatial reduction of monocultures, the installation of ecological compensation areas (e.g., hedgerow network, flowering strips) and a reduction in pesticide use could be appropriate measures.
- The current forests are too dense for some forest species, especially as habitat for grouse. A targeted opening of the forest stands in suitable areas should improve the situation for these species.
- Due to the extensification of use and abandonment in alpine pastures, the forest is spreading. In mountain regions with low elevation amplitude, many alpine species are increasingly losing their habitat. Financial support of alpine pasture management in important population core zones should be considered.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2023.162375>.

CRedit authorship contribution statement

AA, ET, HA, JR, LG and UT conceived the study, BU and ET analysed the data, ET prepared the figures, BU and ET wrote the paper, and all authors commented on the manuscript.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Erich Tasser reports financial support was provided by Division for the Promotion of Education, Universities and Research.

Acknowledgements

The research leading to these results has received funding from the Division for the Promotion of Education, Universities and Research of the Autonomous Province of Bozen/Bolzano - South Tyrol within the project 'Spatio-temporal analysis of the number of hunted animals for the assessment of the landscape quality and the provision of ecosystem services'. We are grateful to the collaborators of the Office for Hunting and Fishery and the FUST-Tirol e.V. for providing data and their expert input. In addition, our thanks and appreciation to the 40 gamekeepers in South Tyrol for their valuable input for interpretation of game species habitat requirements during the questioning. J.R., and U.T. are members of the 'Mountain Regions' research area at the Universität Innsbruck.

References

- Baines, D., Moss, R., Dugan, D., 2004. Capercaillie breeding success in relation to forest habitat and predator abundance: capercaillie breeding, bilberry and predators. *J. Appl. Ecol.* 41, 59–71. <https://doi.org/10.1111/j.1365-2664.2004.00875.x>.
- Balestrieri, A., Remonti, L., Prigioni, C., 2011. Assessing carnivore diet by faecal samples and stomach contents: a case study with alpine red foxes. *Open Life Sci.* 6, 283–292. <https://doi.org/10.2478/s11535-010-0106-1>.
- Bateman, P.W., Fleming, P.A., 2012. Big city life: carnivores in urban environments: urban carnivores. *J. Zool.* 287, 1–23. <https://doi.org/10.1111/j.1469-7998.2011.00887.x>.
- Bender, L.C., Spencer, R.D., 1999. Estimating elk population size by reconstruction from harvest data and herd ratios. *Wildl. Soc. Bull.* 27, 636–645.
- Benoit, D.M., Jackson, D.A., Chu, C., 2021. Partitioning fish communities into guilds for ecological analyses: an overview of current approaches and future directions. *Can. J. Fish. Aquat. Sci.* 78, 984–993. <https://doi.org/10.1139/cjfas-2020-0455>.
- Beyer, H.L., Gurarie, E., Börger, L., Panzacchi, M., Basille, M., Herfindal, I., Van Moorter, B., R. Lele, S., Matthiopoulos, J., 2016. 'You shall not pass!': quantifying barrier permeability and proximity avoidance by animals. *J. Anim. Ecol.* 85, 43–53. <https://doi.org/10.1111/1365-2656.12275>.
- Bradshaw, R.H.W., Hannon, G.E., Lister, A.M., 2003. A long-term perspective on ungulate-vegetation interactions. *For. Ecol. Manag.* 181, 267–280. [https://doi.org/10.1016/S0378-1127\(03\)00138-5](https://doi.org/10.1016/S0378-1127(03)00138-5).
- Cattadori, I.M., Haydon, D.T., Thirgood, S.J., Hudson, P.J., 2003. Are indirect measures of abundance a useful index of population density? The case of red grouse harvesting. *Oikos* 100, 439–446.
- Červinka, J., Drahníková, L., Kreisinger, J., Šálek, M., 2014. Effect of habitat characteristics on mesocarnivore occurrence in urban environment in the Central Europe. *Urban Ecosyst.* 17, 893–909. <https://doi.org/10.1007/s11252-014-0364-1>.
- Chamberlain, D.E., Negro, M., Caprio, E., Rolando, A., 2013. Assessing the sensitivity of alpine birds to potential future changes in habitat and climate to inform management strategies. *Biol. Conserv.* 167, 127–135. <https://doi.org/10.1016/j.biocon.2013.07.036>.
- Clergeau, P., Jokimäki, J., Savard, J.-P.L., 2002. Are urban bird communities influenced by the bird diversity of adjacent landscapes?: urban bird diversity and landscape context. *J. Appl. Ecol.* 38, 1122–1134. <https://doi.org/10.1046/j.1365-2664.2001.00666.x>.
- Corlatti, L., Gugiatti, A., Pedrotti, L., 2016. Spring spotlight counts provide reliable indices to track changes in population size of mountain-dwelling red deer *Cervus elaphus*. *Wildl. Biol.* 22, 268–276. <https://doi.org/10.2981/wlb.00244>.
- Davidson, N.C., 2014. How much wetland has the world lost? Long-term and recent trends in global wetland area. *Mar. Freshw. Res.* 65, 934. <https://doi.org/10.1071/MF14173>.
- Davies, K.F., Melbourne, B.A., James, C.D., Cunningham, R.B., 2010. Using traits of species to understand responses to land use change: birds and livestock grazing in the Australian arid zone. *Biol. Conserv.* 143, 78–85. <https://doi.org/10.1016/j.biocon.2009.09.006>.
- Dumont, L., Lauer, E., Zimmermann, S., Roche, P., Auliac, P., Sarasa, M., 2019. Monitoring black grouse *Tetrao tetrix* in Isère, northern French Alps: cofactors, population trends and potential biases. *Anim. Biodivers. Conserv.*, 227–244. <https://doi.org/10.32800/abc.2019.42.0227>.
- Erhard, H., 2006. Die Entwicklung des Jagdrechts im südlichen Teil Tirols vom 19. bis Anfang des 21. Jahrhunderts. *Schlern* 7, pp. 1–13.
- European Commission, 2020. EU Biodiversity Strategy for 2030. Bringing Nature Back Into Our Lives. Communication for the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions. European Commission.
- European Commission. Directorate General for Environment, 2021. The state of nature in the EU :conservation status and trends of species and habitats protected by the EU nature directives 2013–2018. Publications Office, LU.
- European Environment Agency, 2016. European Forest Ecosystems State and Trends. EUR-OP, Luxembourg.

- Evans, D., 2012. Building the European Union's Natura 2000 network. *Nat. Conserv.* 1, 11–26. <https://doi.org/10.3897/natureconservation.1.1808>.
- Fankhauser, R., Enggist, P., 2004. Simulation of alpine chamois *Rupicapra r. rupicapra* habitat use. *Ecol. Model.* 175, 291–302. <https://doi.org/10.1016/j.ecolmodel.2003.10.020>.
- Ferreira, C., Paupério, J., Alves, P.C., 2010. The usefulness of field data and hunting statistics in the assessment of wild rabbit (*Oryctolagus cuniculus*) conservation status in Portugal. *Wildl. Res.* 37, 223. <https://doi.org/10.1071/WR09137>.
- Foster, D.R., Motzkin, G., Bernardos, D., Cardoza, J., 2002. Wildlife dynamics in the changing New England landscape. *J. Biogeogr.* 29, 1337–1357. <https://doi.org/10.1046/j.1365-2699.2002.00759.x>.
- Frylestam, B., 1980. Reproduction in the European hare in southern Sweden. *Holarct. Ecol.* 3, 74–80.
- Fukasawa, K., Osada, Y., Iijima, H., 2020. Is harvest size a valid indirect measure of abundance for evaluating the population size of game animals using harvest-based estimation? *Wildl. Biol.* 2020, 1–7. <https://doi.org/10.2981/wlb.00708>.
- Furrer, R., Schaub, M., Bossert, A., Isler, R., Jenny, H., Jonas, T., Marti, C., Jenni, L., 2016. Variable decline of alpine rock ptarmigan (*Lagopus muta helvetica*) in Switzerland between regions and sites. *J. Ornithol.* 157, 787–796.
- Galluzzi, M., Armanini, M., Ferrari, G., Zibordi, F., Nocentini, S., Mustoni, A., 2017. Habitat suitability models, for ecological study of the alpine marmot in the central Italian Alps. *Ecol. Inform.* 37, 10–17. <https://doi.org/10.1016/j.ecoinf.2016.11.010>.
- Garcia-Gonzalez, R., Cuartas, P., 1996. Trophic utilization of a montane/subalpine forest by chamois (*Rupicapra pyrenaica*) in the Central Pyrenees. *For. Ecol. Manag.* 88, 15–23. [https://doi.org/10.1016/S0378-1127\(96\)03805-4](https://doi.org/10.1016/S0378-1127(96)03805-4).
- Gianpasquale, C., Alberto, M., 2019. The occurrence and density of three sympatric corvids in a Mediterranean agroecosystem explained by land use. *J. Ornithol.* 160, 1133–1150. <https://doi.org/10.1007/s10336-019-01679-2>.
- Gregory, R.D., van Strien, A., Vorisek, P., Gmelig Meyling, A.W., Noble, D.G., Foppen, R.P.B., Gibbons, D.W., 2005. Developing indicators for European birds. *Philos. Trans. R. Soc. B Biol. Sci.* 360, 269–288. <https://doi.org/10.1098/rstb.2004.1602>.
- Guetté, A., Gaüzère, P., Devictor, V., Jiguet, F., Godet, L., 2017. Measuring the synanthropy of species and communities to monitor the effects of urbanization on biodiversity. *Ecol. Indic.* 79, 139–154. <https://doi.org/10.1016/j.ecolind.2017.04.018>.
- Handels- und Gewerbekammer Bozen, 1894. *Statistischer Bericht der Handels- und Gewerbekammer in Bozen an das k.k. hohe Ministerium für Handel, Gewerbe und öffentliche Bauten: für das Jahr 1890 beziehungsweise für das Quinquennium 1886–1890. Handels- und Gewerbekammer Bozen.*
- Hanley, T.A., 1996. Potential role of deer (Cervidae) as ecological indicators of forest management. *For. Ecol. Manag.* 88, 199–204. [https://doi.org/10.1016/S0378-1127\(96\)03803-0](https://doi.org/10.1016/S0378-1127(96)03803-0).
- Hanley, T.A., 1993. Balancing economic development, biological conservation, and human culture: the Sitka black-tailed deer *Odocoileus hemionus sitkensis* as an ecological indicator. *Biol. Conserv.* 66, 61–67. [https://doi.org/10.1016/0006-3207\(93\)90135-N](https://doi.org/10.1016/0006-3207(93)90135-N).
- Hermoso, V., Carvalho, S.B., Giakoumi, S., Goldsborough, D., Katsanevakis, S., Leontiou, S., Markantonatou, V., Rumes, B., Vogiatzakis, I.N., Yates, K.L., 2022. The EU Biodiversity Strategy for 2030: opportunities and challenges on the path towards biodiversity recovery. *Environ. Sci. Policy* 127, 263–271. <https://doi.org/10.1016/j.envsci.2021.10.028>.
- Heward, C.J., Hoodless, A.N., Conway, G.J., Fuller, R.J., MacColl, A.D.C., Aebischer, N.J., 2018. Habitat correlates of Eurasian woodcock *Scolopax rusticola* abundance in a declining resident population. *J. Ornithol.* 159, 955–965. <https://doi.org/10.1007/s10336-018-1570-z>.
- Hilpold, A., Seeber, J., Fontana, V., Niedrist, G., Rief, A., Steinwandter, M., Tasser, E., Tappeiner, U., 2018a. Decline of rare and specialist species across multiple taxonomic groups after grassland intensification and abandonment. *Biodivers. Conserv.* 27, 3729–3744. <https://doi.org/10.1007/s10531-018-1623-x>.
- Hilpold, A., Seeber, J., Fontana, V., Niedrist, G., Rief, A., Steinwandter, M., Tasser, E., Tappeiner, U., 2018b. Decline of rare and specialist species across multiple taxonomic groups after grassland intensification and abandonment. *Biodivers. Conserv.* 27, 3729–3744.
- Hinojosa, L., Tasser, E., Rüdisser, J., Leitingner, G., Schermer, M., Lambin, E.F., Tappeiner, U., 2019. Geographical heterogeneity in mountain grasslands dynamics in the Austrian-Italian Tyrol region. *Appl. Geogr.* 106, 50–59. <https://doi.org/10.1016/j.apgeog.2019.03.006>.
- Hulbert, I.A.R., Iason, G.R., 1996. The possible effects of landscape change on diet composition and body weight of mountain hares *Lepus timidus*. *Wildl. Biol.* 2, 269–273. <https://doi.org/10.2981/wlb.1996.030>.
- Imperio, S., Bionda, R., Viterbi, R., Provenza, A., 2013. Climate change and human disturbance can lead to local extinction of alpine rock ptarmigan: new insight from the Western Italian Alps. *PLoS ONE* 8, e81598.
- Imperio, S., Ferrante, M., Grignetti, A., Santini, G., Focardi, S., 2010. Investigating population dynamics in ungulates: do hunting statistics make up a good index of population abundance? *Wildl. Biol.* 16, 205–214. <https://doi.org/10.2981/08-051>.
- Jędrzejewska, B., Jędrzejewski, W., Bunevich, A.N., Miłkowski, L., Krasinski, Z.A., 1997. Factors shaping population densities and increase rates of ungulates in Białowieża Primeval Forest (Poland and Belarus) in the 19th and 20th centuries. *Acta Theriol. (Warsz.)* 42, 399–451. <https://doi.org/10.4098/AT.arch.97-39>.
- Johnson, M.D., 2007. Measuring habitat quality: a review. *Condor* 109, 489–504. <https://doi.org/10.1093/condor/109.3.489>.
- Kamieniarz, R., Voigt, U., Panek, M., Strauss, E., Niewiegłowski, H., 2013. The effect of landscape structure on the distribution of brown hare *Lepus europaeus* in farmlands of Germany and Poland. *Acta Theriol. (Warsz.)* 58, 39–46. <https://doi.org/10.1007/s13364-012-0091-z>.
- K.K. Ackerbau-Ministerium, 1896. *Statistisches Jahrbuch: Produktion aus der Seiden- und Bienenzucht in den Jahren 1880, Wildabschuss, Wildschadenvergütung, Torfproduktion in den Jahren 1895. K.K. Hof- und Staatsdruckerei, Wien.*
- K.K. Ackerbau-Ministerium, 1893. *Statistisches Jahrbuch: Produktion aus der Seiden- und Bienenzucht in den Jahren 1885 bis 1891, Wildabschuss, Wildschadenvergütung, Torfproduktion in den Jahren 1886 bis 1891. K.K. Hof- und Staatsdruckerei, Wien.*
- K.K. Ackerbau-Ministerium, 1881. *Statistisches Jahrbuch: Produktion aus der Seiden- und Bienenzucht in den Jahren 1880, Wildabschuss, Wildschadenvergütung, Torfproduktion in den Jahren 1880. Faesy & Frick, Wien.*
- K.K. Ackerbau-Ministerium, 1876. *Statistisches Jahrbuch: Produktion aus der Seiden- und Bienenzucht in den Jahren 1875, Wildabschuss, Wildschadenvergütung, Torfproduktion in den Jahren 1875. Faesy & Frick, Wien.*
- Klansek, E., 2002. Effects of landscape structure on *Perdix perdix* in Austria; outline and outlook. *Z. Für Jagdwiss.* 48, 340–345. <https://doi.org/10.1007/BF02192426>.
- Körner, C., 2012. *Alpine Treelines: Functional Ecology of the Global High Elevation Tree Limits*. Springer, Basel London.
- Lehikoinen, A., Rintala, J., Lammi, E., Pöysä, H., 2016. Habitat-specific population trajectories in boreal waterbirds: alarming trends and bioindicators for wetlands: decline of waterbirds in the boreal eutrophic wetlands. *Anim. Conserv.* 19, 88–95. <https://doi.org/10.1111/acv.12226>.
- Linnell, J.D.C., Cretois, B., Nilsen, E.B., Rolandsen, C.M., Solberg, E.J., Veiberg, V., Kaczensky, P., Van Moorter, B., Panzacchi, M., Rauset, G.R., Kaltenborn, B., 2020. The challenges and opportunities of coexisting with wild ungulates in the human-dominated landscapes of Europe's Anthropocene. *Biol. Conserv.* 244, 108500. <https://doi.org/10.1016/j.biocon.2020.108500>.
- Liu, Y., McCullagh, A., Nieuwenhuis, M., 2018. What factors affect national-scale deer population dynamics in the Republic of Ireland? *Scand. J. For. Res.* 33, 535–549. <https://doi.org/10.1080/02827581.2018.1441902>.
- Maceda-Veiga, A., López, R., Green, A.J., 2017. Dramatic impact of alien carp *Cyprinus carpio* on globally threatened diving ducks and other waterbirds in Mediterranean shallow lakes. *Biol. Conserv.* 212, 74–85. <https://doi.org/10.1016/j.biocon.2017.06.002>.
- Marini, L., Klimke, S., Battisti, A., 2011. Mitigating the impacts of the decline of traditional farming on mountain landscapes and biodiversity: a case study in the European Alps. *Environ. Sci. Policy* 14, 258–267. <https://doi.org/10.1016/j.envsci.2010.12.003>.
- Martinoli, Alessio, Preatoni, D.G., Bisi, F., Gagliardi, A., Martinoli, Adriano, 2017. Where is the pulse to have the finger on? A retrospective analysis of two decades of Alpine Galliforms (Aves: Galliformes) census and game bag data in Italy. *Eur. J. Wildl. Res.* 63, 65. <https://doi.org/10.1007/s10344-017-1122-5>.
- Mayer, A.C., Huovinen, C., 2007. Silvopastoralism in the Alps: native plant species selection under different grazing pressure. *Ecol. Eng.* 29, 372–381. <https://doi.org/10.1016/j.ecoleng.2006.09.015>.
- Mikoláš, M., Svítok, M., Tejkal, M., Leitão, P.J., Morrissey, R.C., Svoboda, M., Seedre, M., Fontaine, J.B., 2015. Evaluating forest management intensity on an umbrella species: Capercaillie persistence in central Europe. *For. Ecol. Manag.* 354, 26–34. <https://doi.org/10.1016/j.foreco.2015.07.001>.
- Musilová, Z., Musil, P., Zouhar, J., Adam, M., 2018. Changes in habitat suitability influence non-breeding distribution of waterbirds in central Europe. *Ibis* 160, 582–596. <https://doi.org/10.1111/ibi.12559>.
- Niedrist, G., Tasser, E., Lüth, C., Dalla Via, J., Tappeiner, U., 2009. Plant diversity declines with recent land use changes in European Alps. *Plant Ecol.* 202, 195–210. <https://doi.org/10.1007/s11258-008-9487-x>.
- Novoa, C., Desmet, J.-F., Brenot, J.-F., Muffat-Joly, B., Arvin-Bérod, M., Resseguier, J., Tran, B., 2019. Chapter twenty. Demographic traits of two alpine populations of rock ptarmigan. In: Sandercock, B.K., Martin, K., Segelbacher, G. (Eds.), *Ecology, Conservation, and Management of Grouse*. University of California Press, pp. 267–280. <https://doi.org/10.1525/9780520950573-022>.
- Obber, F., Celva, R., Libanora, M., Da Rold, G., Dellamaria, D., Partel, P., Ferraro, E., Calabrese, M.S., Morpurgo, L., Pisano, S.R.R., Citterio, C.V., Cassini, R., 2022. Description of a sarcoptic mange outbreak in alpine chamois using an enhanced surveillance approach. *Animals* 12, 2077. <https://doi.org/10.3390/ani12162077>.
- Ozoliņš, J., Pilāts, V., 1995. Distribution and status of small and medium-sized carnivores in Latvia. *Ann. Zool. Fenn.* 32, 21–29.
- Patterson, L., Phelan, J., Goudreau, C., Dykes, R., 2017. Flow-biology relationships based on fish habitat guilds in North Carolina. *JAWRA J. Am. Water Resour. Assoc.* 53, 56–66. <https://doi.org/10.1111/1752-1688.12495>.
- Pereboom, V., Mergey, M., Villerette, N., Helder, R., Gerard, J.-F., Lodé, T., 2008. Movement patterns, habitat selection, and corridor use of a typical woodland-dweller species, the European pine marten (*Martes martes*), in fragmented landscape. *Can. J. Zool.* 86, 983–991. <https://doi.org/10.1139/Z08-076>.
- Poudel, B.S., Spooner, P.G., Matthews, A., 2015. Temporal shift in activity patterns of Himalayan marmots in relation to pastoralism. *Behav. Ecol.* 26, 1345–1351. <https://doi.org/10.1093/beheco/arv083>.
- Prigioni, C., Balestrieri, A., Remonti, L., Cavada, L., 2008. Differential use of food and habitat by sympatric carnivores in the eastern Italian Alps. *Ital. J. Zool.* 75, 173–184. <https://doi.org/10.1080/11250000701885521>.
- Ramousse, R., Métal, J., Le Berre, M., 2009. Twenty-seventh year of the Alpine marmot introduction in the agricultural landscape of the Central Massif (France). *Ethol. Ecol. Evol.* 21, 243–250. <https://doi.org/10.1080/08927014.2009.9522479>.
- Ranta, E., Lindström, J., Lindén, H., Helle, P., 2008. How reliable are harvesting data for analyses of spatio-temporal population dynamics? *Oikos* 117, 1461–1468. <https://doi.org/10.1111/j.0030-1299.2008.16879.x>.
- Rehnus, M., 2013. *Der Schneehase in den Alpen: ein Überlebenskünstler mit ungewisser Zukunft, Bristol-Schriftenreihe. Haupt [u.a.], Bern.*
- Rehnus, M., Bollmann, K., Schmatz, D.R., Hackländer, K., Braunisch, V., 2018. Alpine glacial relict species losing out to climate change: the case of the fragmented mountain hare population (*Lepus timidus*) in the Alps. *Glob. Chang. Biol.* 24, 3236–3253. <https://doi.org/10.1111/gcb.14087>.
- Rehnus, M., Wehrle, M., Palme, R., 2014. Mountain hares *Lepus timidus* and tourism: stress events and reactions. *J. Appl. Ecol.* 51, 6–12. <https://doi.org/10.1111/1365-2664.12174>.
- Rödel, H.G., Dekker, J.J.A., 2012. Influence of weather factors on population dynamics of two lagomorph species based on hunting bag records. *Eur. J. Wildl. Res.* 58, 923–932. <https://doi.org/10.1007/s10344-012-0635-1>.

- Rodewald, A.D., Gehrt, S.D., 2014. Wildlife population dynamics in urban landscapes. In: McCleery, R.A., Moorman, C.E., Peterson, M.N. (Eds.), *Urban Wildlife*. Springer, US, Boston, MA, pp. 117–147 https://doi.org/10.1007/978-1-4899-7500-3_8.
- Roeder, N., Lederbogen, D., Trautner, J., Bergamini, A., Stofer, S., Scheidegger, C., 2010. The impact of changing agricultural policies on jointly used rough pastures in the bavarian pre-Alps: an economic and ecological scenario approach. *Ecol. Econ.* 69, 2435–2447. <https://doi.org/10.1016/j.ecolecon.2010.07.013>.
- Rüdiger, J., Tasser, E., Peham, T., Meyer, E., Tappeiner, U., 2015. The dark side of biodiversity: spatial application of the biological soil quality indicator (BSQ). *Ecol. Indic.* 53, 240–246. <https://doi.org/10.1016/j.ecolind.2015.02.006>.
- Schäublin, S., Bollmann, K., 2011. Winter habitat selection and conservation of hazel grouse (*Bonasa bonasia*) in mountain forests. *J. Ornithol.* 152, 179–192. <https://doi.org/10.1007/s10336-010-0563-3>.
- Schemnitz, S.D., Batcheller, G.R., Matthew, J., White, H.B., Fall, M.W., 2012. Capturing and handling wild animals. In: Silvy, N.J. (Ed.), *The Wildlife Techniques Manual*. Johns Hopkins University Press, Baltimore, Md, pp. 232–269.
- Schmieder, K., Werner, S., Bauer, H.-G., 2006. Submersed macrophytes as a food source for wintering waterbirds at Lake Constance. *Aquat. Bot.* 84, 245–250. <https://doi.org/10.1016/j.aquabot.2005.09.006>.
- Scholefield, P., Firbank, L., Butler, S., Norris, K., Jones, L.M., Petit, S., 2011. Modelling the European farmland bird indicator in response to forecast land-use change in Europe. *Ecol. Indic.* 11, 46–51. <https://doi.org/10.1016/j.ecolind.2009.09.008>.
- Schulz, T., 2015. An uphill struggle against scrub encroachment: implementation of the alpine pasturing subsidy scheme in Switzerland. *Land Use Policy* 42, 318–328. <https://doi.org/10.1016/j.landusepol.2014.08.001>.
- Schweiger, A.-K., Nopp-Mayr, U., Zohmann, M., 2012. Small-scale habitat use of black grouse (*Tetrao tetrix* L.) and rock ptarmigan (*Lagopus muta helvetica* Thienemann) in the Austrian Alps. *Eur. J. Wildl. Res.* 58, 35–45. <https://doi.org/10.1007/s10344-011-0537-7>.
- Seidl, R., Albrich, K., Erb, K., Formayer, H., Leidinger, D., Leitinger, G., Tappeiner, U., Tasser, E., Rammer, W., 2019. What drives the future supply of regulating ecosystem services in a mountain forest landscape? *For. Ecol. Manag.* 445, 37–47. <https://doi.org/10.1016/j.foreco.2019.03.047>.
- Shochat, E., Lerman, S., Fernández-Juricic, E., 2015. Birds in urban ecosystems: population dynamics, community structure, biodiversity, and conservation. In: Aitkenhead-Peterson, J., Volder, A. (Eds.), *Agronomy Monographs*. American Society of Agronomy, Crop Science Society of America, Soil Science Society of America, Madison, WI, USA, pp. 75–86 <https://doi.org/10.2134/agronmonogr55.c4>.
- Soininen, E.M., Fuglei, E., Pedersen, Å.Ø., 2016. Complementary use of density estimates and hunting statistics: different sides of the same story? *Eur. J. Wildl. Res.* 62, 151–160. <https://doi.org/10.1007/s10344-016-0987-z>.
- Suchant, R., Baritz, R., Braunisch, V., 2003. Wildlife habitat analysis – a multidimensional habitat management model. *J. Nat. Conserv.* 10, 253–268. <https://doi.org/10.1078/1617-1381-00026>.
- Summers, R.W., Green, R.E., Proctor, R., Dugan, D., Lambie, D., Moncrieff, R., Moss, R., Baines, D., 2004. An experimental study of the effects of predation on the breeding productivity of capercaillie and black grouse. *J. Appl. Ecol.* 41, 513–525. <https://doi.org/10.1111/j.0021-8901.2004.00891.x>.
- Tappeiner, U., Borsdorf, A., Tasser, E., 2008. *Alpenatlas Atlas des Alpes = Atlante delle Alpi = Atlas Alp = Mapping the Alps: society, economy, environment*. Spektrum Akademischer Verlag, Heidelberg.
- Tasser, E., Leitinger, G., Tappeiner, U., 2017. Climate change versus land-use change—what affects the mountain landscapes more? *Land Use Policy* 60, 60–72.
- Tasser, E., Ruffini, F.V., Tappeiner, U., 2009. An integrative approach for analysing landscape dynamics in diverse cultivated and natural mountain areas. *Landsc. Ecol.* 24, 611–628. <https://doi.org/10.1007/s10980-009-9337-9>.
- Tryjanowski, P., Morelli, F., Mikula, P., Krištin, A., Indykiewicz, P., Grzywaczewski, G., Kronenberg, J., Jerzak, L., 2017. Bird diversity in urban green space: a large-scale analysis of differences between parks and cemeteries in Central Europe. *Urban For. Urban Green.* 27, 264–271. <https://doi.org/10.1016/j.ufug.2017.08.014>.
- Unterholzner, L., 2017. *Atlas der Brutvögel Südtirols*. Arbeitsgemeinschaft für Vogelkunde, Meran.
- Unterthurner, B., Agreiter, A., Gerstgrasser, L., Hochrainer, S., Rüdiger, J., Tappeiner, U., Tasser, E., 2019. *Landschaft im Visier. Ein Überblick über die historische Entwicklung der jagdbaren Tierarten in Südtirol*. EURAC research, Bozen.
- Vierikko, K., Pellikka, J., Hanski, I.K., Myllyviita, T., Niemelä, J., Vehkamäki, S., Lindén, H., 2010. Indicators of sustainable forestry: the association between wildlife species and forest structure in Finland. *Ecol. Indic.* 10, 361–369. <https://doi.org/10.1016/j.ecolind.2009.06.019>.